

Argand diagrams

Basic skills

Q1. a) z is at $(2, 4)$ $z = x + yi$
so $z = 2 + 4i$

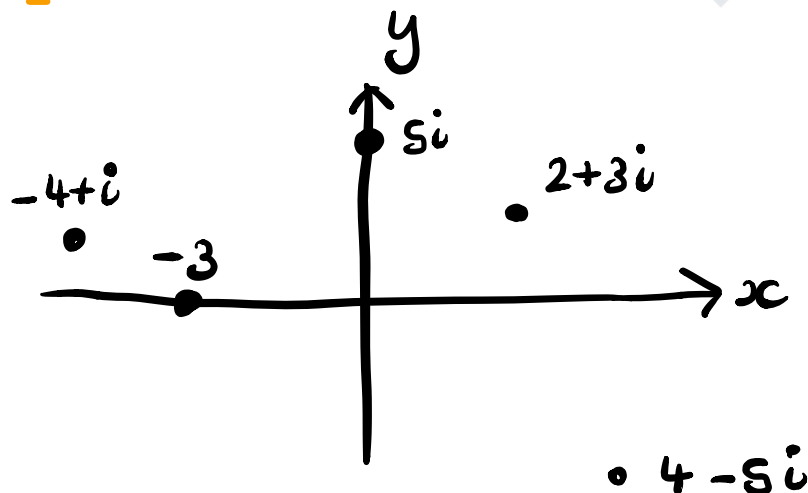
b) w is at $(6, -4)$
 $w = 6 - 4i$

c) u is at $(0, -6)$
 $u = 0 - 6i = -6i$

d) v is at $(-6, 1)$
 $v = -6 + i$

Competency skills

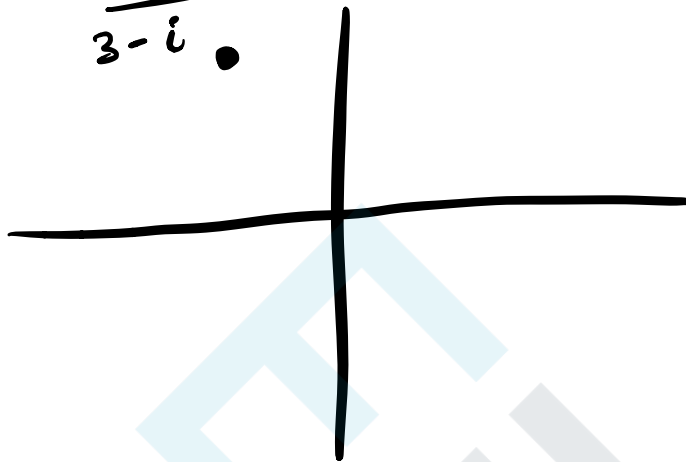
Q2.



Q3. a) $\frac{4i}{3-i} \times \frac{3+i}{3+i} = \frac{12i-4}{9+1} = \frac{12i-4}{10}$

$$= -\frac{4}{10} + \frac{12}{10}i$$

b) $\frac{4i}{3-i} \cdot$



Q4. $z = 2+i$, $w = 3i-4$

First calculate

c) $z-w = (2+i) - (3i-4) = 6-2i$

d) $3z+2w = 3(2+i) + 2(3i-4) = -2+9i$

e) $z^* = 2-i$

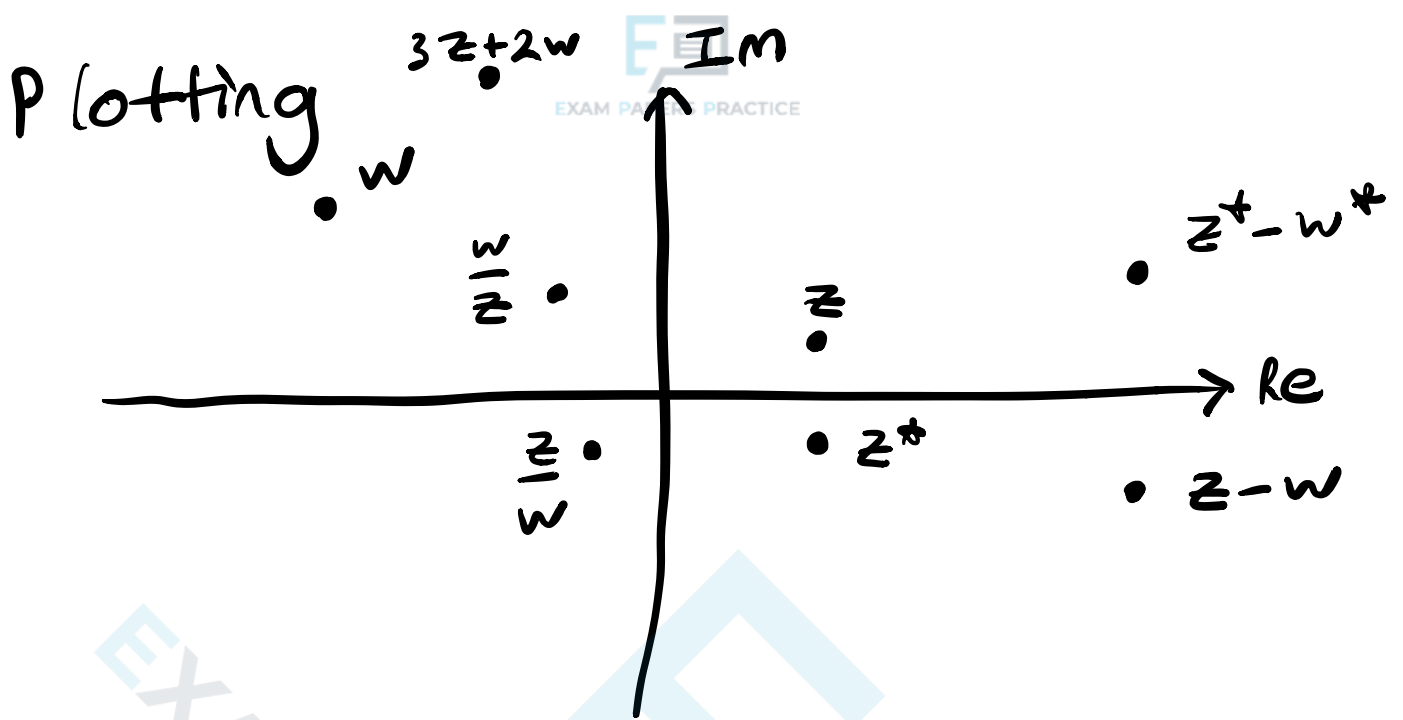
f) $z^*-w^* = (2-i) - (-3i-4) = 6+2i$

g) $\frac{z}{w} = \frac{2+i}{3i-4} \times \frac{-3i-4}{-3i-4} = \frac{-8+3-4i-6i}{9+16}$

$$= -\frac{5}{25} - \frac{10}{25}i = -\frac{1}{5} - \frac{2}{5}i$$

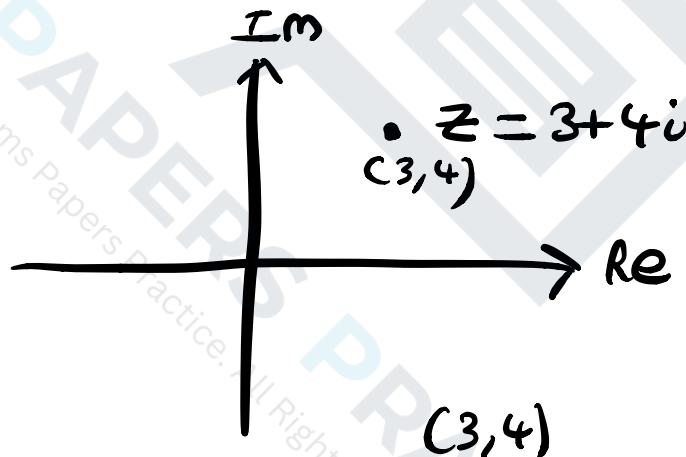
h) $\frac{w}{z} = \frac{3i-4}{2+i} \times \frac{2-i}{2-i} = \frac{-8+3+4i+6i}{4+1} = \frac{-5+10i}{5}$

$$= -1+2i$$

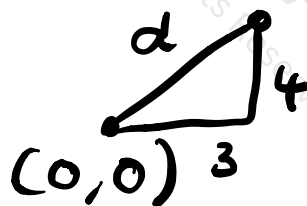


Advanced skills

Qs. a)



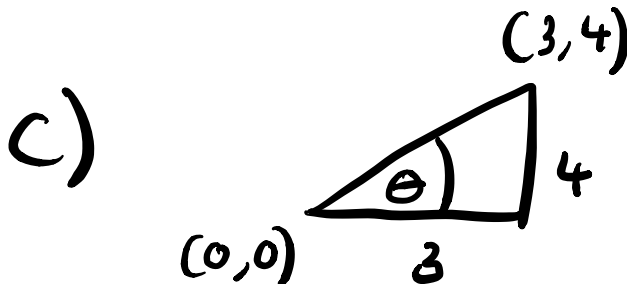
b)



so

$$d^2 = 3^2 + 4^2$$

$$d = \sqrt{9 + 16} = 5$$



$$\tan \theta = \frac{4}{3}$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right)$$

d) so 5 away at 0.927

$$= 0.927..$$

i.e $x = 5 \cos(0.927) + i 5 \sin(0.927)$

Q6. a) If $b^2 - 4ac > 0$ then 2 real solutions so both roots on Real / x -axis

e.g.

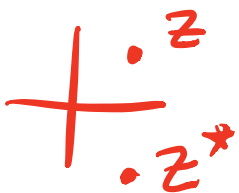


b) If one is a reflection in the y -axis of the other

x -axis is reflect is $\rightarrow$ then it is not complex

(as z, z^* solutions are reflection in x -axis)

so $d \geq 0$



complex conjugates